CSE 585 Recap

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Agenda

- Recap
- Status
- Debate

Semester in Review

Resource-Centric View

- Datacenter as a Computer
  - Scale-out architecture but has all the same components as a single machine
  - High parallelism, diverse workloads, heterogeneous resources, failures, and communication-driven performance

- GPU as a Computer
  - Scale-up architecture but has all the same components as a single machine
  - High parallelism, diverse workloads, heterogeneous resources, failures, and communication-driven performance (at least for LLMs)

Common "Systems" Techniques

Technique	Performance/Efficiency	Availability/Resilience
Replication & Erasure coding	X	X
Sharding/partitioning	X	X
Scheduling & Load balancing	X	
Health & Integrity checks		X
Compression & Quantization	X	
Centralized controller	X	
Canaries		X
Speculation & Redundant execution	X	

Workload-Centric View

- Basics
- Pre-Training
- Post-Training
- Inference
- Agentic Systems
- Hardware/Infrastructure
- Power and Energy

Basics

Transformers

- Quadratic self-attention forms the backbone of modern LLMs
- Compute-bound in training, but memory (bandwidth)-bound during inference

Training

- Different kinds of parallelism: data, tensor, and pipeline parallelism
- How to schedule them in an optimized fashion (1F1B)

LLM Inference

- Prefill vs. Decode
- Continuous batching
- KVCache and its memory management

Pre-Training (8 Papers)

- How to manage memory?
- How to manage large sequences?
- How to automatically find better plans for execution?
- How to increase hardware and software reliability?
- How to recover from failures?

Post-Training (4 Papers)

- What’s the tradeoff between scaling compute vs scaling model parameters?
- How to use reinforcement learning effectively?
- How to optimize a synchronous RLHF framework?
- How to optimize an asynchronous one?

Inference (10 Papers)

- How to optimize throughput and latency of serving? Can we do both simultaneously?
- How to balance compute and communication in inference end-to-end?
- How to serve long-context inputs?
- How to server multimodal inputs?
- How to serve in resource-constrained setups?
- What are the other metrics? How do we pick a metric?
- How to evaluate systems?

Agentic Systems (4 Papers)

- How to optimize agentic applications?
- How to optimize external data sources for inference (RAG)?

Hardware + Power/Energy (6 Papers)

- How to vertically integrate a model with the hardware?
- How to optimize the cost when building/managing a datacenter?
- How to reduce the energy and power consumption of training and inference?
- What's the impact of power fluctuations and how to reduce them?

Ethics

- We must carefully consider MANY aspects when evaluating AI and AI agents
  - Societal
  - Environmental
  - Financial
  - Etc.

- There is no right answer, but less wrong ones

Papers We Dislike

Status

Grades

Remaining Steps

- Poster presentation on Dec 3 (Wednesday) 10-12 at Tishman
  - Email us final project titles by Noon on Dec 2 (Tuesday)
  - See https://edstem.org/us/courses/80826/discussion/7359492
- Research paper
  - Write like an ICML paper with ICML template and page constraints
  - As if you'd submit it to a workshop with ~3 more months of work or to a conference after ~6 more months of work
  - Read How to Write a Great Research Paper by Simon Peyton Jones
  - Due on or before 1:00PM EST on Dec 15

Thank You!

Please complete the course evaluation

Let's Debate!